DEV CON VIII INDIA

Decoupled Consensus



Mumbai L1 R&D Workshop

Breakout · Part 1 · 2 November 2026 · Room 205A

Same page today.

- Newest developments
- Open questions
- Your concerns

Together all week.



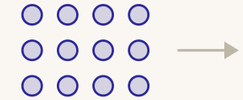
Working on DC? Let’s meet.

Telegram: DC breakouts · any free room

Steps ahead


2022 – 25
Research
Oct 2025
Cambridge
Not BFT-only
Nov 2025
Devconnect ARG
First designs
Jan 2026
Berlin
Strawmap
Mar 2026
ethresear.ch
DC public
Oct 2026
Cambridge
Workshop
21 Oct 2026
EIP-8453
2 Nov 2026
Mumbai
L1 R&D Workshop
We are here

Why we need decoupled consensus



committee PQ signatures, as they are

Post-quantum readiness

Small committee, PQ signatures as they are



Fixing Gasper

Reorg-resilient chain, separate from finality

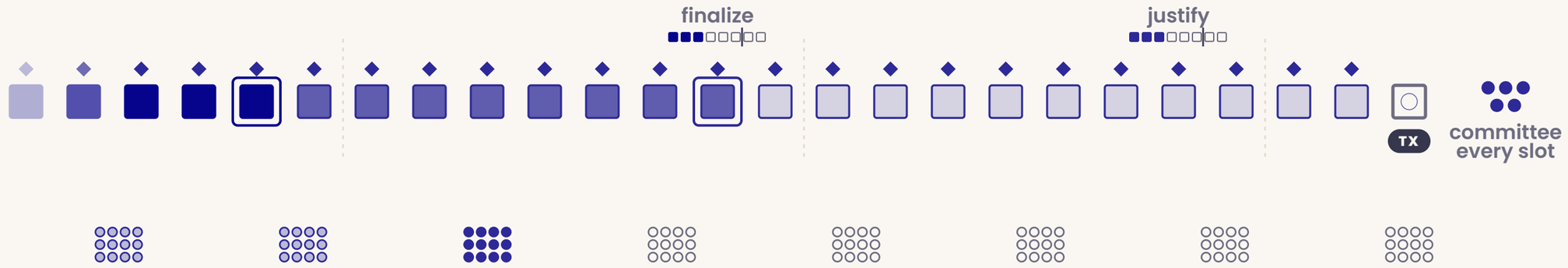


Fast finality, quick slots

At least 4× faster, off the critical path

Decoupled consensus, as it stands

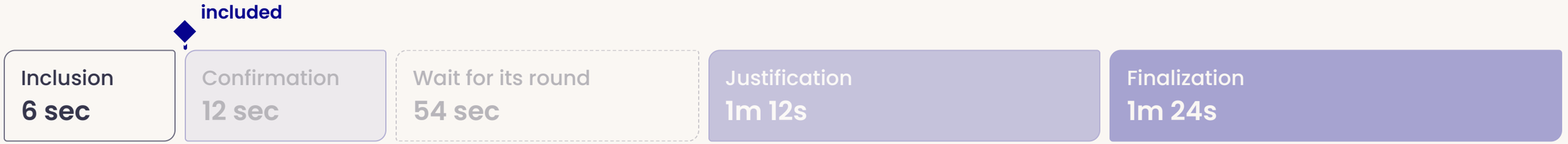
Available chain · Goldfish with locks



Finality gadget · full validator set, one cohort per slot

One transaction, inclusion to finality

~3m 48s expected · today ~16 min



Expected values · finality.watch model · 8-slot rounds · ~1.1M validators · 12-second slots

DC lays the rails.

NEXT TRAIN

L* · ONE-ROUND FINALITY

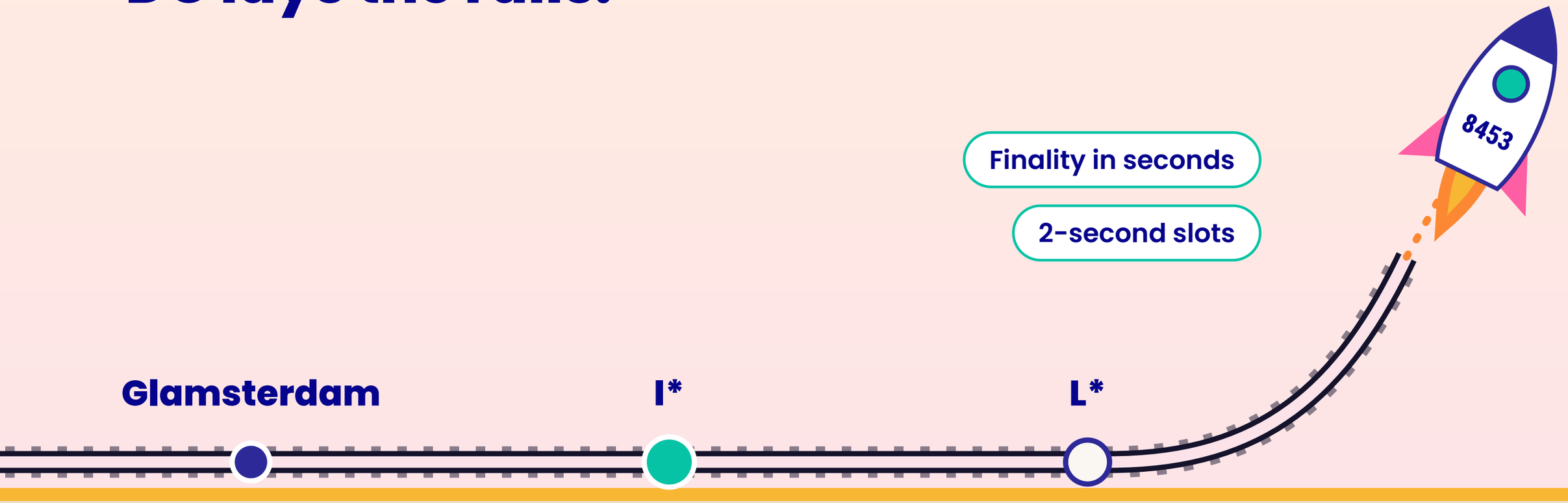


4× faster finality | optimal inactivity leak | PQ heartbeat | accountable safety | height progress | committee self-selection

dynamic availability | validator anonymity | 50% fault tolerance | minority veto | healing | secret leader election | lean state

leak fairness | perfect reward fairness | just one more property

DC lays the rails.



Fast confirmation

Decoupled consensus

One-round finality

Today: the core of each layer

Available chain

Safe confirmations

Safe honest proposals

Finality gadget

Accountable safety

No honest slashing

Finality liveness

Hardening

Bounded asynchrony

Healing

Incentives

Leak fairness

Incentive compatible

Networking

Small change

Forward compatible

Uses the whole round

Degrades gracefully

This afternoon

13:30

Why DC	Yann
Goals	**Roberto**
Goldfish	Francesco
Networking	Sukun & Yann

15:15

Specs	Mikhail
Prototyping	Terence
Q&A	

Appendix

THE GATEWAY OF I*

YOU SHALL NOT PASS

4× faster finality

one-round finality

accountable safety

quick slots

dynamic availability

PQ readiness

optimal inactivity leak

healing

height progress

leak fairness

just one more property

small network change



We can have it all.

Just not all at once.

Ben, product owner of DC

Was there 3,000 years ago

Phase 0, in crypto years

What do we say to the god of scope creep?



Not in I*.

There's always 1RF.



I WAS THERE, 3,000 YEARS AGO



“JUST ONE MORE PROPERTY”
SOUNDED REASONABLE THEN TOO